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Hierarchically aggregated tri-stage network for visible-depth-thermal salient object detection

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HIGHLIGHTS

- A tri-stage visible-depth-thermal network aligns, fuses, and refines saliency cues for salient object detection.
- CRE and PEGI reduce cross-modal mismatches in the channel and spatial domains.
- AMCA and CGF strengthen cross-scale aggregation for complex objects.
- HATNet outperforms 17 methods on VDT-2048, especially in hard scenes.

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ABSTRACT

Visible-depth-thermal salient object detection leverages complementary cues from three modalities to identify visually prominent objects, which generally requires cross-modal alignment and boundary refinement before and after multimodal aggregation, respectively. Existing methods tend to jointly handle alignment, aggregation, and refinement within a monolithic pipeline, which entangles their conflicting optimization objectives and limits performance. To address these issues, we propose a hierarchically aggregated tri-stage network (HATNet) that decouples these objectives into three progressive stages. Specifically, the alignment stage employs a channel reweighting exchange (CRE) module for channel-level feature alignment, and a position-enriched graph-based interaction (PEGI) module to establish global spatial correspondence, thereby mitigating cross-modal feature conflicts. The merging stage employs an adaptive multimodal convolution aggregation (AMCA) module for same-scale feature fusion and a context guidance fusion (CGF) module for cross-scale feature aggregation via bidirectional attention. The purification stage employs a selective feature refinement (SFR) module to enhance object boundaries and semantic consistency. Extensive experiments on the VDT-2048 dataset demonstrate that HATNet consistently outperforms 17 state-of-the-art models across multiple evaluation metrics, particularly in challenging scenarios with complex backgrounds and cross-scale objects.

1. Introduction

Salient object detection (SOD) aims to identify and segment the most visually prominent objects in images, which serves as a crucial preprocessing step for various computer vision downstream tasks [1–3], such as object detection in diverse application domains [4] and abnormal behavior detection in video surveillance [5]. Traditional SOD methods primarily rely on a single visible-light modality for the extraction of visual features (e.g., color, texture, and shape) [6]. Early approaches focused on discriminative regional feature integration to leverage visual features, while more advanced architectures integrated fusion, feedback, and focus mechanisms to enhance detection performance.

However, relying solely on visual cues often proves insufficient for single-modal SOD, particularly in challenging scenarios involving cluttered backgrounds, low illumination, and adverse environmental conditions [7,8]. This performance limitation [9–11] has motivated researchers to explore multimodal fusion approaches by incorporating complementary depth (D) and thermal (T) modalities. As illustrated in Fig. 1, the integration of multiple modalities demonstrates superior performance compared to single-modal approaches in challenging scenarios. Depth maps provide geometric spatial information between objects and the camera, effectively enhancing foreground–background segmentation in cluttered scenes, though they are susceptible to sensor

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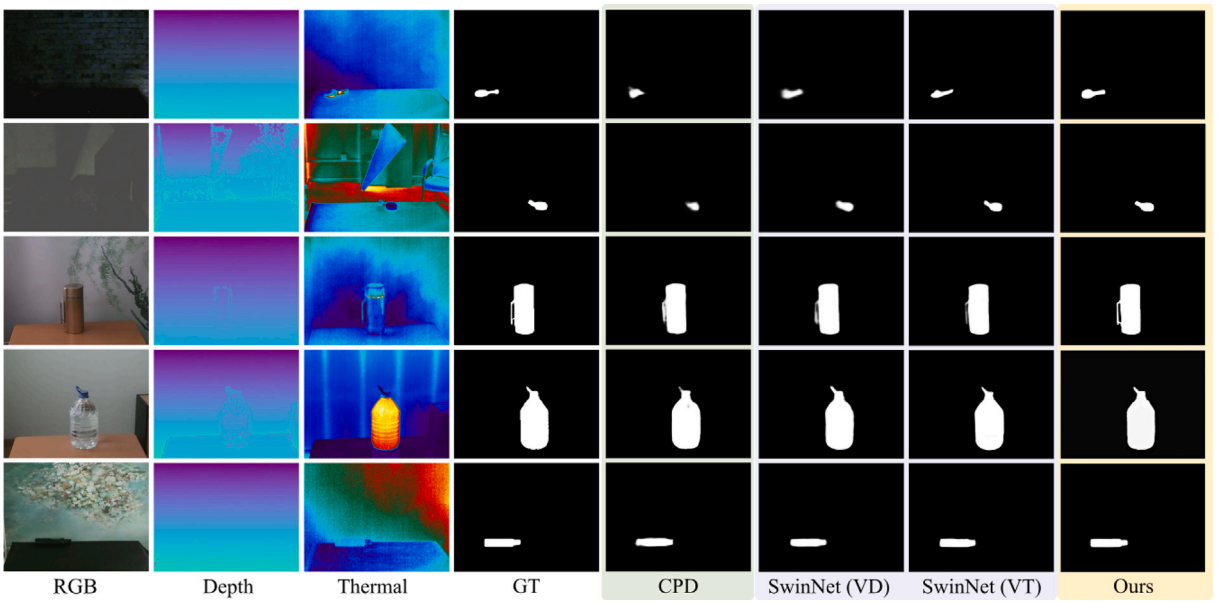


Fig. 1. Comparison of single-modal and multimodal methods under challenging scenarios. CPD uses visible images only, SwinNet uses dual-modal images, and ours adopts visible-depth-thermal triple-modal images.

noise, range attenuation, and planar overlap issues [12–16]. Thermal images excel in low-light conditions but may generate misleading cues in high-temperature crossover scenarios, where background-target thermal radiation confusion occurs due to similar thermal signatures between objects and their surroundings [8,17,18]. Since dual-modal methods have limitations in individual use yet possess complementary features, researchers have begun to explore approaches for visible-depth-thermal SOD [19–21].

Early triple-modal SOD methods mostly lacked modality-specific procedures in the feature fusion stage. Such one-size-fits-all approaches not only fail to fully leverage the complementarity of each modality, but also tend to incur feature conflicts, both in which restrict the further improvement of model performance. Recent years have witnessed the emergence of cross-modal interaction strategies based on adaptive weight allocation and attention mechanism guidance, which balance modality-specific contributions and reduce conflicts and redundancy in feature fusion.

Despite progress, current methods still face three intertwined difficulties. Before fusion, the task requires them to align cross-modal features, but modal heterogeneity often causes distribution mismatches [11,16,22,23]. During fusion, the task requires them to aggregate multi-scale information, yet cross-scale dependencies are often inadequately captured [24–26]. After fusion, the task requires them to refine object boundaries, whereas boundary optimization and semantic consistency are rarely coordinated [2,7,26,27]. These three aspects are typically addressed within a conflated pipeline, without accounting for the fact that alignment, aggregation, and refinement have inherently conflicting optimization objectives. To illustrate, smoothing for semantic integrity tends to blur edges, sharpening for boundaries amplifies noise, and denoising inevitably suppresses useful high-frequency cues. Such conflicts limit the performance of existing VDT-SOD methods.

This paper introduces HATNet, a hierarchical tri-stage network designed to address these issues by progressively integrating multimodal information through alignment, merging, and purification. HATNet effectively bridges cross-modal discrepancies using CRE-PEGI modules, adaptively captures features across different scales with AMCA and CGF, and improves boundary accuracy through SFR in the spatial-frequency coupling domain. The main contributions can be concluded as follows:

- A structured framework named hierarchically aggregated tri-stage network (HATNet) is proposed to decouple modal alignment, semantic aggregation, and boundary refinement into three progressive stages for visible-depth-thermal SOD tasks.
- We develop a channel reweighting exchange (CRE) module and a position-enriched graph-based interaction (PEGI) module to mitigate the modal heterogeneity that causes misaligned feature distributions in the alignment stage. The former calibrates channel-level semantic discrepancies, while the latter establishes spatially-constrained global correspondence through geometric priors.
- We design an adaptive multimodal convolution aggregation (AMCA) module and a context guidance fusion (CGF) module to bridge the semantic gap between pixel-level fusion and high-level saliency prediction in the merging stage. The former dynamically captures cross-scale targets, while the latter enables semantic-guided cross-scale propagation.
- We deploy a selective feature refinement (SFR) module to jointly optimize boundary sharpness and structural consistency in the purification stage, which synergizes cross-scale spatial correlations with frequency-domain context for coordinated adjustment.

2. Related work

2.1. Single-modal salient object detection

The development of single-modal salient object detection (SOD) has primarily focused on enhancing feature modeling capabilities. CNN-based methods achieved strong feature extraction through regional integration, fusion mechanisms, and global context modeling [2,6,28,29]. However, CNNs are constrained by their fixed receptive fields, which restrict the ability to capture long-range semantic dependencies. In response to this limitation, Transformer-based approaches emerged to leverage global context modeling [22], though they have often underperformed in boundary precision due to inadequate perception of details. Recent research has focused on multi-scale feature fusion and boundary refinement through strategies such as pyramidal attention and foundation model enhancement [3,10,30]. Additionally, weakly supervised approaches have been explored due to their reduced dependence on expensive labeled data [31,32]. Recent studies have also extended

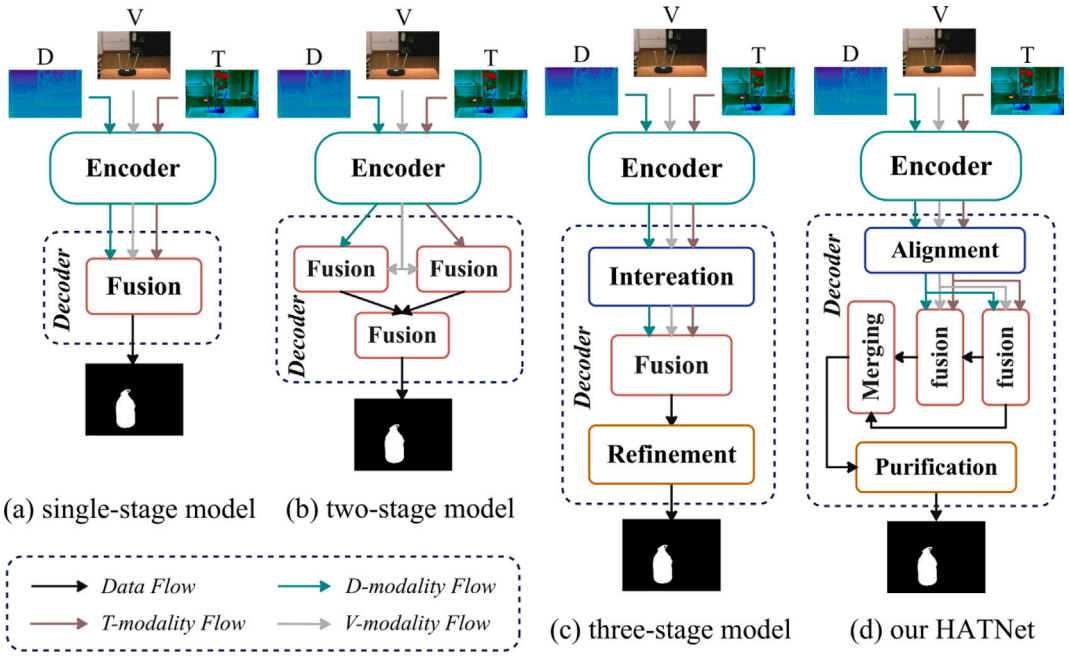


Fig. 2. Comparison of existing triple-modal SOD pipeline structures. (a) one-stage model directly fuses multimodal features. (b) two-stage model separates fusion and refinement. (c) tri-stage model adds intermediate fusion. (d) Our HATNet adopts hierarchical alignment and purification across multiple stages.

SOD to specialized imaging scenarios through maximum-parallax modeling for light-field images and interactive edge awareness for optical remote sensing images [33,34].

2.2. Dual-modal salient object detection

While single-modal SOD has achieved substantial progress, it struggles in extreme environments. Researchers have thus incorporated depth (RGB-D) or thermal (RGB-T) data to complement visible-light imagery. In RGB-D SOD, a primary research direction is feature fusion, where attention mechanisms have been widely adopted. Representative works such as DPANet [14], which induces gated attention to adjust fusion weights, and subsequent methods [11,13,15,16,35] have developed strategies to select complementary features and localize salient regions. Recently, Transformer architectures have been adopted, including DFormerv2 [36] which uses depth maps as geometric priors and EM-Trans [22] which incorporates edge-aware mechanisms to preserve boundary details. RGB-T research targets scenarios such as low-light conditions where robust fusion is essential. Recent *Applied Soft Computing* studies have further emphasized robust cross-modal interaction under adverse illumination, including frequency-guided saliency modeling [37], infrared-visible fusion with skip-connecting group convolutional attention [38], and security-oriented infrared-visible fusion under high-level visual tasks [39]. A large-scale benchmark for RGB-T SOD was introduced in [18], while enhanced detection networks were developed in [40,41].

As research in the field has advanced, the community has increasingly shifted dual-modal SOD toward unified frameworks [42]. For instance, unified information fusion with a shared backbone and adaptive fusion module was proposed in [43], and a saliency prototype framework supporting flexible multi-task fusion was established in [44]. Furthermore, several Transformer-based approaches have emerged: EM-Trans enhanced RGB-depth alignment through edge-aware mechanisms [22], CompTr provided a general complementary Transformer paradigm across dense prediction tasks [45], and SwinNet further demonstrated the efficacy of edge-aware architectures for cross-modal tasks [46].

2.3. Triple-modal salient object detection

Triple-modal frameworks use visible light, depth, and thermal infrared modalities for better robustness and scene understanding. Early research primarily concentrated on dataset construction and foundational fusion frameworks. The first triple-modal salient object detection network, HWSI, together with the VDT-2048 dataset, was introduced in [20]; MFFNet was subsequently developed to explore complementary features through encoder integration and multi-scale fusion [21]; and hierarchical two-stage modal fusion for triple-modal saliency prediction was further explored in [47]. However, these methods were limited by simple feature concatenation or shallow fusion strategies that hindered their ability to capture deep semantic correlations between modalities. Subsequent research has focused on refined feature fusion strategies. QSFNet, a quality-aware fusion network that dynamically selects reliable regions based on quality evaluation, was introduced in [19] to effectively mitigate environmental interference. Despite these advancements, challenges remain in modality misalignment during feature encoding.

Recent advancements have focused on enhanced modality interaction and innovative architectures. TMNet introduced triple-modal interaction encoders and cross-modal attention mechanisms [24]. IFENet proposed an interaction-fusion-enhancement mechanism [48], while dynamic weighted fusion [27], feature integration [23], and lightweight three-stream decoding [25] further improved triple-modal fusion. Despite these innovations, current methods still struggle with cross-modal feature alignment, multi-scale semantic aggregation, and coordinated boundary refinement, which hinder further improvements in detection accuracy and robustness.

3. The proposed method

3.1. Network overview

Previous research adopts one-stage, two-stage, or tri-stage architectures [19,20] as shown in Fig. 2(a)–(c). Nevertheless, single- or two-stage designs fail to account for the trilemma in optimization objectives among the three sub-tasks of noise suppression, semantic aggregation, and boundary refinement. To elaborate, smoothing for

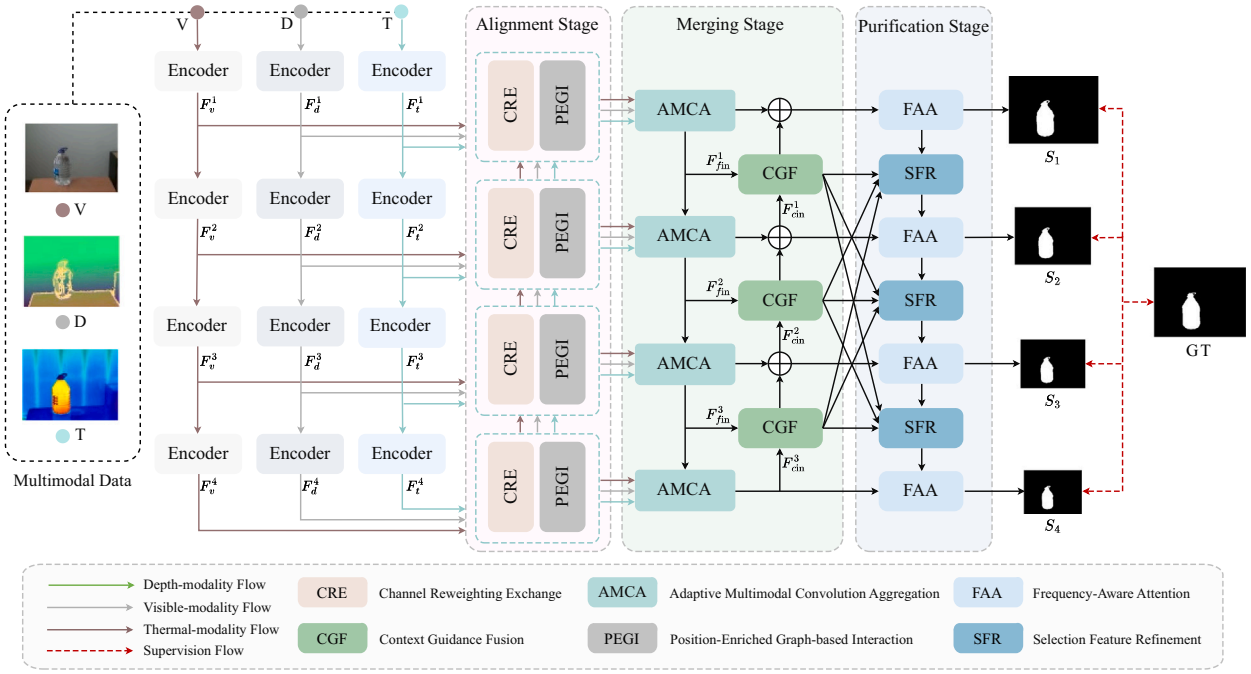


Fig. 3. An overview of the proposed HATNet. Our model is designed as a tri-stage network that progressively refines features from three different modalities. The alignment, merging, and purification stages respectively correspond to feature alignment, multi-scale fusion, and boundary refinement to produce the final saliency map.

semantic integrity inevitably blurs edges, sharpening for boundaries amplifies noise, and denoising suppresses useful high-frequency cues. These inherent conflicts prevent single- or two-stage designs from achieving a globally optimal solution.

Moreover, recently proposed tri-stage models (i.e., IFENet [48]) are distinguished by their use of linear processing mechanisms. While these models offer more refinement than one-stage and two-stage networks, their performance in dense prediction tasks in computer vision is hindered by a limited capacity for cross-scale feature exploration. To this end, we propose HATNet, a hierarchical tri-stage framework depicted in Fig. 2(d), which coordinates multimodal fusion and multi-scale aggregation while preserving salient-object boundaries under modal heterogeneity.

Our three-stage architecture decouples noise suppression, semantic aggregation, and boundary refinement into separate stages, which not only circumvents the optimization trilemma but also allows us to predefine the information flow in each stage through inductive priors. Such fine-grained optimization enables us to outperform conventional multi-scale fusion work, as validated by our experiments in Section 4.1.

As illustrated in Fig. 3, HATNet features a hierarchical pipeline with three progressive stages. For an input of spatial size $H \times W$, a weight-shared backbone extracts four scale-specific feature maps with spatial resolutions of $\frac{H}{4} \times \frac{W}{4}$, $\frac{H}{8} \times \frac{W}{8}$, $\frac{H}{16} \times \frac{W}{16}$, and $\frac{H}{32} \times \frac{W}{32}$, respectively. In the first stage, channel reweighting exchange (CRE) and position-enriched graph-based interaction (PEGI) align same-scale features across the three modalities. The two modules independently compute cross-modally aligned representations at each scale. The second stage uses adaptive multimodal convolution aggregation (AMCA) and context guidance fusion (CGF) for coarse-to-fine propagation. High-level semantics from deeper layers guide low-level spatial details across adjacent scales. In the third stage, selective feature refinement (SFR), supported by frequency-aware aggregation (FAA), refines the multi-scale features. Cross-connections exchange complementary information across scales before the network produces multi-scale saliency predictions that receive ground-truth supervision.

3.2. Alignment stage

To eliminate cross-modal distribution discrepancies prior to fusion, we introduce the alignment stage after backbone feature extraction. This stage jointly introduces channel reweighting exchange (CRE) and position-enriched graph-based interaction (PEGI) modules, supplemented by equivariant regularization loss, to align multimodal feature distributions. Inspired by recent adaptive feature refinement techniques [49], the alignment stage resolves local feature conflicts before multimodal fusion. The alignment stage operates as a channel-spatial collaborative module, where CRE reweights each modality's channels based on cross-modal similarity to enhance complementary signals and suppress inconsistent ones, while PEGI models semantic and geometric relationships to ensure spatially aligned cross-modal interactions.

Modality-specific priors cause multimodal feature maps extracted from backbone networks to exhibit significant differences in channel-level semantic composition; the CRE module coordinates modality-specific representations so as to address such disparities. Channel-level complementarity is enhanced while conflicts are mitigated before spatial reasoning occurs, which ultimately aligns cross-modal semantic subspaces. Given three input features $\mathbf{F}_v, \mathbf{F}_d, \mathbf{F}_t \in \mathbb{R}^{C\times H\times W}$, the method first normalizes and embeds these features into a unified channel space. Depthwise separable convolution then processes the normalized features to capture local inter-channel dependencies:

$$\mathbf{F}_m^{\mathrm{DW}} = \mathrm{DWConv}(\mathrm{Linear}(\mathrm{PConv}(\mathrm{LN}(\mathbf{F}_m)))) \tag{1}$$

where $m \in \{v, d, t\}$. The operation enhances local channel discriminability and suppresses irrelevant noise. For each modal pair, cosine similarity between channel embeddings produces attention weights that recalibrate each channel distribution (Fig. 4).

While CRE provides semantic alignment in the channel domain, global cross-modal spatial alignment remains a significant challenge [22,36]. Geometric misalignment can occur between features from different modalities for the same target, due to discrepancies in sensor

resolution, perspective, or noise. Additionally, conventional graph-based reasoning methods lack explicit spatial constraints and thus may construct erroneous semantic links between unrelated regions. In response to these structural limitations, PEGI incorporates explicit geometric priors into the graph construction. The model injects spatial attributes into node embeddings and constructs a graph based on the dual constraints of semantic similarity and spatial proximity. This design enables global feature interaction through spatial anchoring and preserves consistency between cross-modal features in both spatial localization and semantic association. For a given feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$, the model generates position-aware features and fuses them with original appearance features to produce enhanced spatial-aware representations:

$$\begin{cases} \mathbf{F}_{\text{geo}} = \text{Concat}(\phi_{\text{pos}}(x,y),\ \phi_{\text{dist}}(x,y), \phi_{\text{angle}}(x,y), \phi_{\text{scale}}(x,y)) \\ \mathbf{F}_{\text{enh}} = \text{ReLU}(\text{BN}(\text{Conv}_{1\times1}([\mathbf{F}, \mathbf{F}_{\text{geo}}]))) \end{cases} \tag{2}$$

Each mapping function ϕ projects normalized spatial descriptors, including coordinate grids, center distances, polar angles, and radial scales, into the feature embedding space. For the pixel-to-node mapping, a soft assignment mechanism $\mathbf{S}$ allocates pixel features to K nodes. The core of this process is the construction of an adjacency matrix $\mathbf{A}$, which integrates semantic similarity and geometric proximity priors as:

$$\mathbf{A} = \lambda \cdot \mathbf{A}_{\text{sem}} + (1-\lambda)\cdot \mathbf{A}_{\text{geo}} \tag{3}$$

where $\mathbf{A}_{\text{sem}}$ is based on the cosine similarity of node semantic features, and $\mathbf{A}_{\text{geo}}$ is determined by the Gaussian distance between node spatial locations. The parameter λ serves to balance the weights of these two priors, and this fusion weight is learnable during training to adaptively adjust the contribution of semantic and geometric constraints. A graph neural network then utilizes the adjacency matrix $\mathbf{A}$ to update the node features to $\mathbf{H}_{\text{node}}$. Finally, these updated node features are projected back to the pixel space:

$$\hat{\mathbf{F}} = \mathbf{F}_{\text{enh}} + R(\mathbf{H}_{\text{node}} \odot \mathbf{S}) \tag{4}$$

Thus, the enhanced features $\hat{\mathbf{F}}$ are formed by combining the original features $\mathbf{F}_{\text{enh}}$ with the graph-structured node information, which accomplishes a spatially-aware feature enhancement. Despite the alignment of local channels and global structures, the feature distributions of multimodal features can still diverge. For instance, deep features, influenced by perspective projection, often form a radial distribution, whereas RGB features tend to create texture-oriented clusters. Such discrepancies in feature space distributions can cause pattern conflicts during feature fusion. To resolve this conflict, an equivariance regularization loss is introduced. This loss aligns the pairwise relationships among cross-modal features within the feature space, compelling the triple-modal features to adhere to a unified implicit distribution. Given the extracted features of each modality as $\{\mathbf{z}_i^v\}, \{\mathbf{z}_i^d\}, \{\mathbf{z}_i^t\}$, the model computes their intra-modal pairwise distance matrices. The optimization objective is an alignment loss designed to minimize the Frobenius distance between these matrices across modalities.

$$\begin{cases} \mathcal{L}_{\text{align}} = \dfrac{\sum_{\substack{x,y\in\{v,d,t\}\\ x\neq y}} \|\mathbf{D}_x - \mathbf{D}_y\|_F}{2} \\ \mathbf{D}_m(i,j) = \|\mathbf{z}_i^m - \mathbf{z}_j^m\|_2 \end{cases} \tag{5}$$

where $\mathbf{D}_m \in \mathbb{R}^{N\times N}$ is the feature distance matrix for modality $m \in \{v,d,t\}$, and $\|\cdot\|_F$ and $\|\cdot\|_2$ denote the Frobenius norm and the 2-norm of a matrix, respectively.

3.3. Merging stage

The merging stage receives multimodal features from the CRE and PEGI alignment modules to achieve deep information fusion and multi-scale feature aggregation. Our approach employs a progressive processing logic that advances from modal fusion to scale aggregation through two designed modules, AMCA and CGF respectively. Initially, the AMCA module performs a coarse fusion of the heterogeneous multimodal features. Subsequently, based on the result of this coarse fusion, the CGF module conducts a contextual aggregation of multi-scale features. The process culminates in a feature representation that embodies both modal complementarity and scale integrity.

Conventional multi-scale fusion treats all modalities symmetrically across different scales. AMCA selects modal responses within each scale, whereas CGF propagates semantic guidance bidirectionally between adjacent scales.

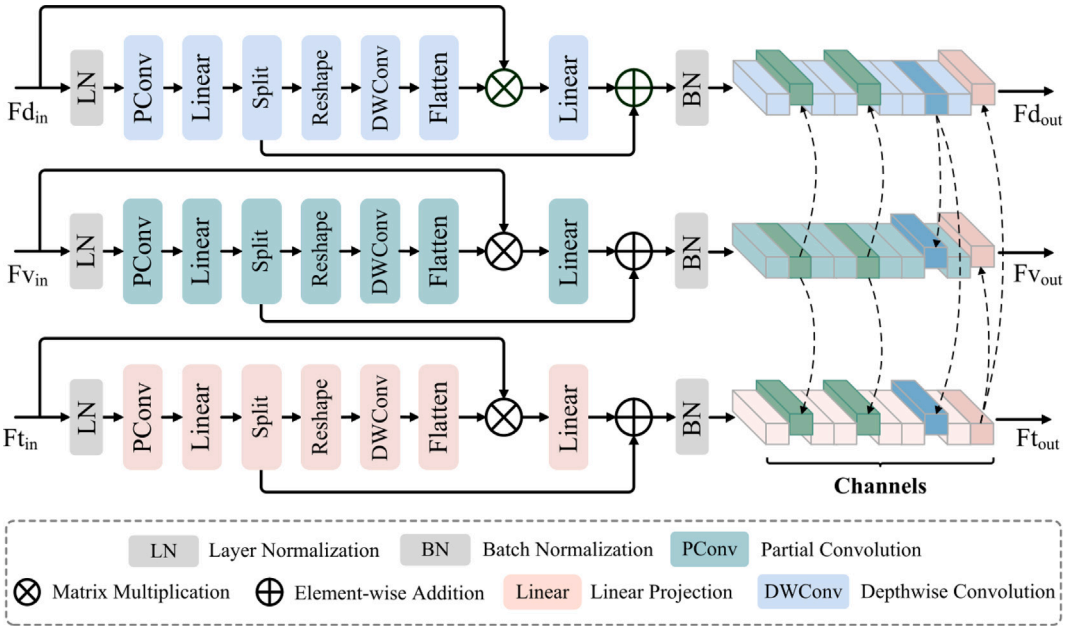


Fig. 4. Workflow of the channel reweighting exchange (CRE) module. Features undergo normalization, group-wise processing, channel reweighting, and cross-modal exchange for enhanced multimodal fusion.

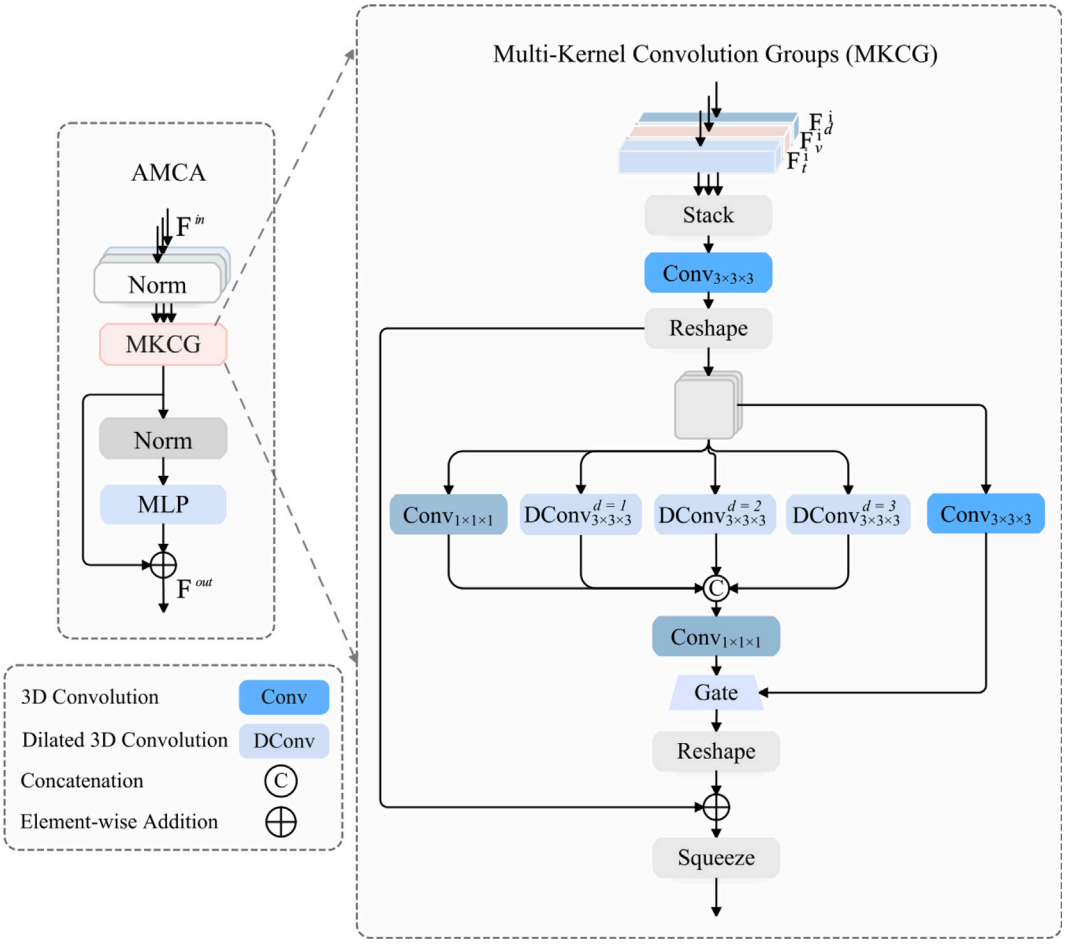


Fig. 5. Workflow of adaptive multimodal convolution aggregation (AMCA) module. This block integrates multimodal features using multi-kernel 3D convolutions and gated attention mechanisms.

The AMCA module focuses on integrating multimodal features at the same scale. Its core objective is to adaptively fuse features for targets of various sizes by fully leveraging multimodal information. Such an approach ensures the capture of local detail correlations in small targets while preserving the global structural information of large ones. Specifically, AMCA receives the aligned multimodal feature tensor F^{in} from the Alignment Stage, which contains visible, depth, and thermal-infrared modalities. Unlike traditional channel concatenation, these modalities are stacked along a new modal dimension to form a four-dimensional fusion input. After normalization, the stacked features are processed through convolution and fed into multiple parallel branches to generate the fused feature:

$$\begin{cases} \mathrm{F}_{\mathrm{conv}}^{1\times1\times1} = \mathrm{Conv}_{1\times1\times1}\left(R\left(\mathrm{Conv}_{3\times3\times3}\left(\mathrm{F}_{\mathrm{st}}\right)\right)\right) \\ \mathrm{F}_{\mathrm{dconv}}^{d} = \mathrm{DConv}_{3\times3\times3}^{d}\left(R\left(\mathrm{Conv}_{3\times3\times3}\left(\mathrm{F}_{\mathrm{st}}\right)\right)\right) \\ \mathrm{F}_{\mathrm{c}} = \mathrm{Conv}_{1\times1\times1}\left(\mathrm{F}_{\mathrm{conv}}^{1\times1}, \mathrm{F}_{\mathrm{dconv}}^{d=1}, \mathrm{F}_{\mathrm{dconv}}^{d=2}, \mathrm{F}_{\mathrm{dconv}}^{d=3}\right) \end{cases} \tag{6}$$

where $d \in \{1, 2, 3\}$ and R denotes the Reshape operation. The generation of gating weights $\boldsymbol{F}_w$ involves the application of a $3 \times 3 \times 3$ convolution to the initial features, followed by a sigmoid activation, to adaptively weight the multi-field-of-view features $\boldsymbol{F}_c$. The weighted features are then reshaped and fused with the initial features through a residual connection. A final squeeze operation produces the output as:

$$\boldsymbol{F}_w = \sigma\left(\mathrm{Conv}_{3\times3\times3}\left(R\left(\mathrm{Conv}_{3\times3\times3}\left(\boldsymbol{F}_{st}\right)\right)\right)\right) \tag{7}$$

For further augmentation of the feature representation, a non-linear transformation is implemented using normalization and a multi-layer perceptron (MLP). The transformation is then combined with the input through a residual connection, which generates the final output of the module as:

$$\boldsymbol{F}^{out} = \boldsymbol{F}^{in} \oplus \mathrm{MLP}\left(\mathrm{Norm}\left(S\left(R\left(\mathrm{Conv}_{3\times3\times3}\left(\boldsymbol{F}_{st}\right)\right) \oplus R\left(\boldsymbol{F}_c \odot \boldsymbol{F}_w\right)\right)\right)\right) \tag{8}$$

where σ, $\odot$, $\oplus$, R, and S denote the sigmoid activation function, element-wise multiplication, the residual connection, the Reshape operation, and the Squeeze operation, respectively (Fig. 5).

Subsequent to the intra-scale feature integration by the AMCA module, the CGF module aggregates features from adjacent scales and thereby achieves cross-scale synergy [24,26]. The input to the CGF module consists of the adjacent-scale features from the AMCA output. A primary challenge stems from the discrepancy in channel dimensions across different scales, as direct interaction would cause a mismatch in feature distributions. To resolve this, the module first utilizes a channel embedding operation that maps both coarse- and fine-grained features to tensors of a uniform dimension. Efficient cross-scale aggregation is then facilitated by a bidirectional attention guidance mechanism, which establishes a bridge between scales through the generation of two distinct guidance maps. The computational logic for both can be uniformly represented as:

$$\begin{cases} \mathbf{G}_{\mathrm{coa}} = \mathrm{Softmax}\left(\dfrac{\mathbf{K}_{\mathrm{coa}}\mathbf{K}_{\mathrm{coa}}^{\top}}{\sqrt{dk}}\right) \odot \mathbf{V}_{\mathrm{coa}} \\ \mathbf{G}_{\mathrm{fin}} = \mathrm{Softmax}\left(\dfrac{\mathbf{K}_{\mathrm{fin}}\mathbf{K}_{\mathrm{fin}}^{\top}}{\sqrt{dk}}\right) \odot \mathbf{V}_{\mathrm{fin}} \end{cases} \tag{9}$$

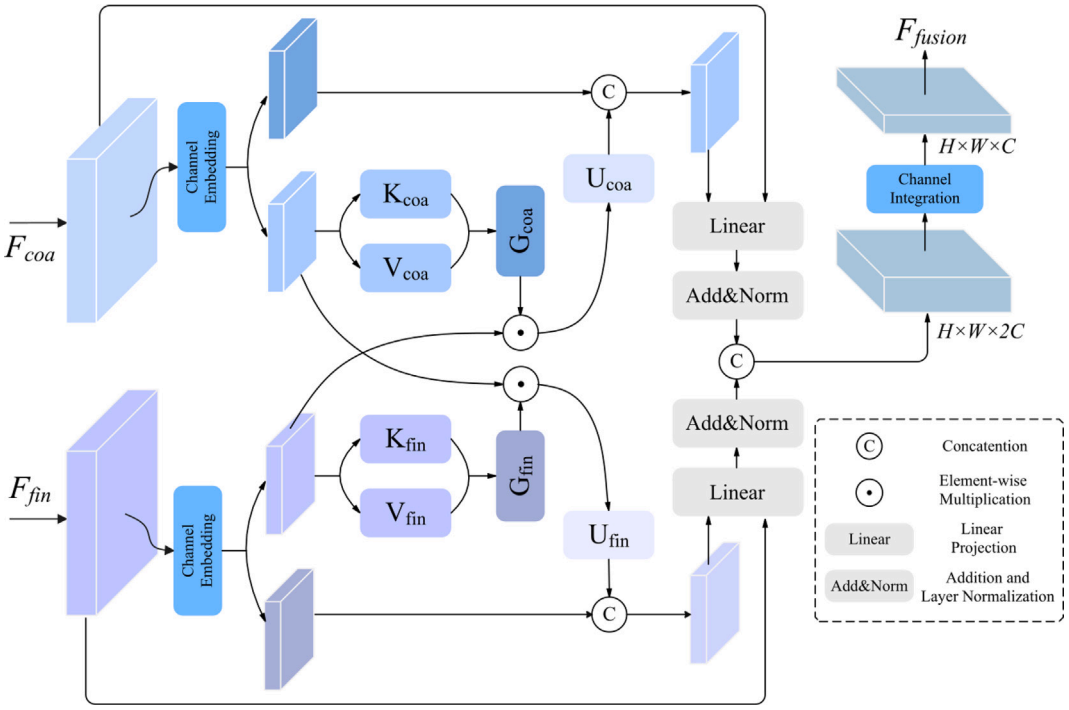


Fig. 6. Workflow of context guidance fusion (CGF) module. This block integrates coarse-grained and fine-grained features through cross-scale attention and residual connections.

where d_k denotes the dimension of the query vector. The guidance map $\mathbf{G}_{\text{coa}}$ encodes the distribution of key high-level semantic regions, whereas $\mathbf{G}_{\text{fin}}$ captures fine-grained spatial details. To enable cross-scale information exchange, an element-wise multiplication is performed between these guidance maps and the features from the opposite scale. Specifically, the multiplication of the coarse-grained guidance map with fine-grained features steers the focus of low-level information toward critical semantic areas. Conversely, the combination of the fine-grained guidance map with coarse-grained features serves to augment the high-level representations with precise local details. This entire procedure is formulated as:

$$\begin{cases} \mathbf{F}_{\text{cr-coa}} = \mathbf{G}_{\text{fin}} \odot \mathbf{F}_{\text{coa}} \\ \mathbf{F}_{\text{cr-fin}} = \mathbf{G}_{\text{coa}} \odot \mathbf{F}_{\text{fin}} \end{cases} \tag{10}$$

where $\mathbf{F}_{\text{coa}}$ and $\mathbf{F}_{\text{fin}}$ represent the coarse and fine-grained features after channel dimension unification, respectively; $\otimes$ denotes element-wise multiplication; and $\mathbf{F}_{\text{cr-coa}}$ and $\mathbf{F}_{\text{cr-fin}}$ are the resulting features after cross-scale interaction.

To preserve fundamental feature information and enhance feature robustness, the module subsequently generates auxiliary features. A linear transformation is applied to both $\mathbf{F}_{\text{cr-coa}}$ and $\mathbf{F}_{\text{cr-fin}}$ to produce auxiliary features $\mathbf{U}'_{\text{coa}}$ and $\mathbf{U}'_{\text{fin}}$, which retain the core information. Following this, a residual connection is established between these auxiliary features and the original input features, and a layer normalization operation is concurrently applied to stabilize the feature distribution.

$$\begin{cases} \mathbf{U}_{\text{coa}}'' = \text{Norm}\left(\mathbf{U}_{\text{coa}}' \oplus \mathbf{F}_{\text{coa}}\right) \\ \mathbf{U}_{\text{fin}}'' = \text{Norm}\left(\mathbf{U}_{\text{fin}}' \oplus \mathbf{F}_{\text{fin}}\right) \end{cases} \tag{11}$$

where $\oplus$ denotes the residual connection and Norm represents the layer normalization operation.

Finally, the auxiliary features $\mathbf{U}_{\text{coa}}''$ and $\mathbf{U}_{\text{fin}}''$, after the residual and normalization steps, are concatenated along the channel dimension. A subsequent channel integration operation serves to compress redundant channels and strengthen the associations among effective features. This process generates the final output feature, which possesses comprehensive multi-scale integrity as:

$$\mathbf{F}_{\text{fus}} = \text{Linear}_{\text{int}}\left(\text{Concat}(\mathbf{U}''_{\text{coa}}, \mathbf{U}''_{\text{fin}})\right) \tag{12}$$

where Concat is the channel concatenation operation and $\text{Linear}_{\text{int}}$ denotes the linear transformation for channel integration. This procedure enables the CGF module to establish a deep association between features of adjacent scales. In synergy with the AMCA module, a progressive modality fusion-scale aggregation process is created, yielding a final feature representation that embodies both multimodal complementarity and multi-scale integrity (Fig. 6).

3.4. Purification stage

Although the alignment and fusion stages achieve a preliminary integration of multimodal features across different modalities and scales, challenges persist in complex scenarios. A notable issue is the disconnection that can occur between the local details of fine-grained features and the semantic information of coarse-grained features during multi-scale interaction. This problem often results in blurred object boundaries or a lack of regional consistency. Consequently, the purification stage introduces a selective feature refinement (SFR) module that performs precise feature purification by deeply exploring cross-scale feature correlations and synergizing with frequency-domain optimized features. Its purpose is to supply high-quality features for the final saliency prediction.

Conventional boundary postprocessing often separates edge enhancement from regional semantic recovery. SFR coordinates cross-scale details with frequency-domain semantics supplied by FAA, thereby preserving semantic consistency and sharpening uncertain object boundaries.

The design of the SFR module balances cross-scale information complementarity with multi-domain feature fusion. The module accepts the multi-scale feature set $\{\mathbf{F}^{o_1}_{\text{cgf}}, \mathbf{F}^{o_2}_{\text{cgf}}, \mathbf{F}^{o_3}_{\text{cgf}}\}$ from the CGF module as input and produces refined features $\mathbf{F}^{o_i}_{\text{sfr}}$. The process commences by capturing implicit cross-scale correlations through global semantic association. Specifically, global average pooling is applied to the input features at each scale to extract channel-wise global statistics. This operation filters

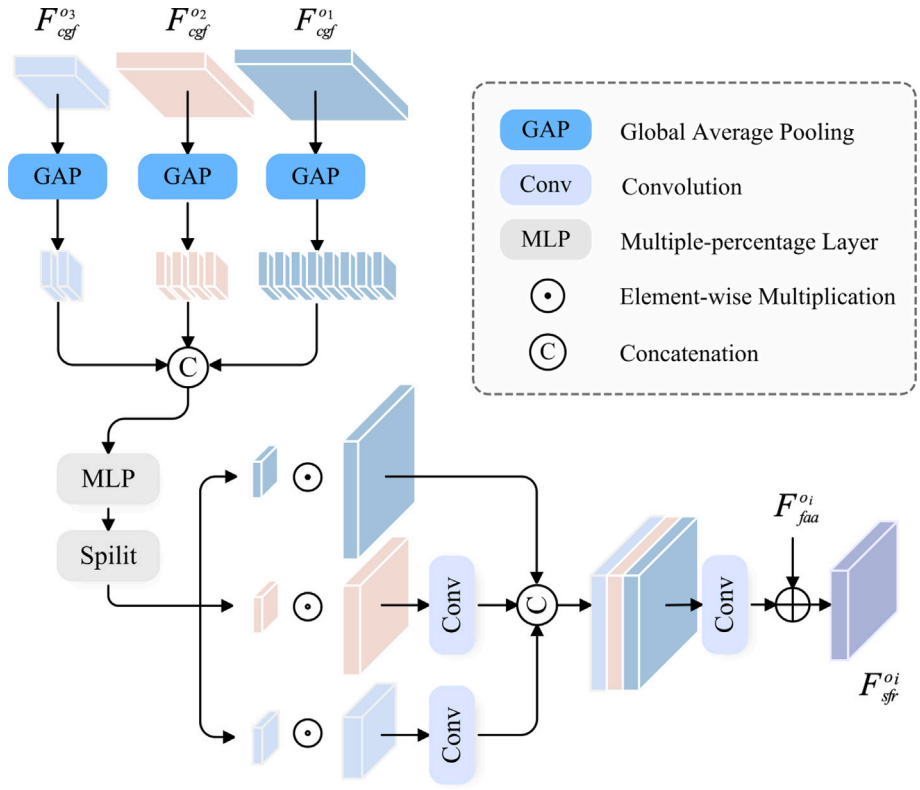


Fig. 7. Workflow of the selective feature refinement (SFR) module. This block purifies features by integrating cross-scale spatial correlations with frequency-domain context.

local noise and focuses on the most discriminative semantic patterns within each scale. The global statistics from all scales are then concatenated into a joint feature vector. A multi-layer perceptron (MLP) performs a non-linear mapping on this vector to generate a global correlation vector $\mathbf{G}$, which models the inter-channel associations across different scales. A subsequent Split operation divides $\mathbf{G}$ into attention weight sub-vectors $\{\mathbf{g}_1, \mathbf{g}_2, \mathbf{g}_3\}$ corresponding to each scale, as follows:

$$\begin{cases} \mathbf{G}^j_{\text{gap}} = \text{GAP}\left(\mathbf{F}^{o_j}_{\text{cgf}}\right) \\ \mathbf{G} = \text{MLP}\left(\text{Concat}\left(\mathbf{G}^1_{\text{gap}}, \mathbf{G}^2_{\text{gap}}, \mathbf{G}^3_{\text{gap}}\right)\right) \\ \{\mathbf{g}_1, \mathbf{g}_2, \mathbf{g}_3\} = \text{Split}(\mathbf{G}) \end{cases} \tag{13}$$

where $j = \{1, 2, 3\}$.

Based on these attention weights, the features of the three scales undergo differential enhancement. For the first two scales, the features are weighted by their respective attention vectors and then processed through a convolutional layer to restore spatial resolution and capture local spatial dependencies. The third-scale feature, being rich in global semantics, is preserved directly after attention weighting. This entire procedure facilitates a continuous cross-scale refinement, where small-scale features leverage large-scale semantics to focus on target regions, and large-scale features are supplemented with small-scale details to enrich local structures. The process is formulated as:

$$\begin{cases} \mathbf{F}^1_{\text{conv}} = \text{Conv}\left(\mathbf{F}^{o_1}_{\text{cgf}} \odot \mathbf{g}_1\right) \\ \mathbf{F}^2_{\text{conv}} = \text{Conv}\left(\mathbf{F}^{o_2}_{\text{cgf}} \odot \mathbf{g}_2\right) \\ \mathbf{F}^3_{\text{raw}} = \mathbf{F}^{o_3}_{\text{cgf}} \odot \mathbf{g}_3 \\ \mathbf{F}_{\text{fus}} = \text{Conv}\left(\text{Concat}\left(\mathbf{F}^1_{\text{conv}}, \mathbf{F}^2_{\text{conv}}, \mathbf{F}^3_{\text{raw}}\right)\right) \\ \mathbf{F}^{o_i}_{\text{sfr}} = \mathbf{F}_{\text{fus}} \oplus \mathbf{F}^{o_i}_{\text{faa}} \end{cases} \tag{14}$$

A particularly noteworthy aspect is the integration of spatial and frequency domains. The frequency-domain optimized features, denoted as $\mathbf{F}^{o_i}_{\text{faa}}$, are obtained by separating high-frequency and low-frequency components. The high-frequency parts supplement boundary details, while the low-frequency parts reinforce semantic associations. Inspired by the frequency-domain processing strategy of the FSE module in IFENet [48], we independently design FAA to aggregate high- and low-frequency components for boundary-detail supplementation and semantic reinforcement. Through a residual connection with the fused features $\mathbf{F}_{\text{fus}}$, a deep complementarity is established between the global correlation information of the spatial domain and the high-frequency details and low-frequency semantics of the frequency domain. From a spatial perspective, the cross-scale refined features, after being optimized in the frequency domain, exhibit sharper boundaries and stronger regional consistency. From a frequency perspective, the frequency-domain features are anchored by cross-scale semantics, which suppresses noise and strengthens salient targets. The final output, $\mathbf{F}^{o_i}_{\text{sfr}}$, thus ensures semantic consistency and detail integrity via cross-scale refinement while also gaining improved noise resistance and feature purity from the spatial-frequency fusion. This synergy establishes a high-quality foundation for saliency prediction, effectively resolving the issue of multi-scale feature disconnection and enabling clear object boundaries with unified regional semantics in complex scenes (Fig. 7).

3.5. Supervision strategy

To align with the tri-stage architecture of HATNet, a phased supervision strategy ensures that the alignment, merging, and purification stages receive effective gradient signals. Specifically, the alignment stage evaluates single-modality features through unimodal predictions. The merging stage assesses cross-modal aggregation through multi-scale saliency predictions. Finally, the purification stage evaluates feature refinement through boundary and body predictions.

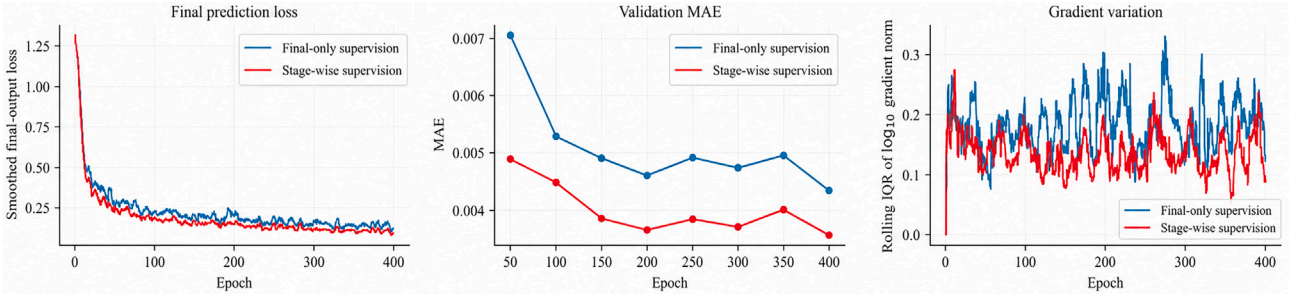


Fig. 8. Training dynamics under stage-wise and final-only supervision.

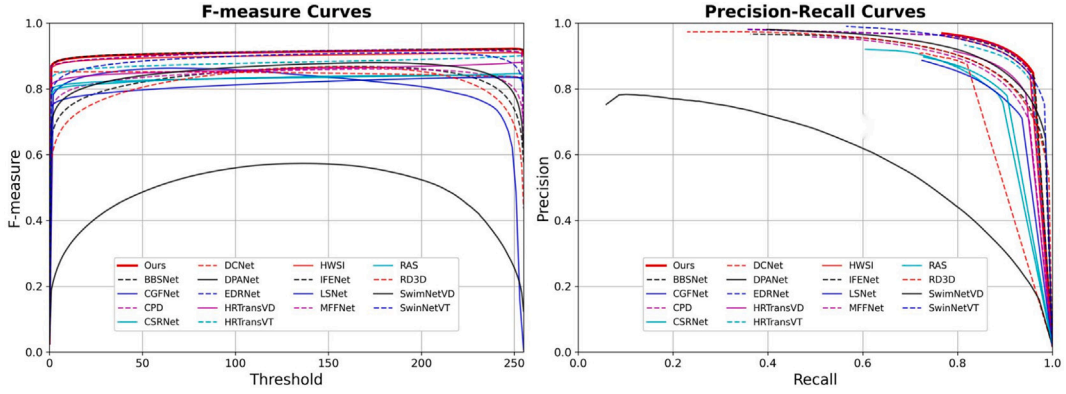


Fig. 9. Quantitative evaluation of different saliency models on the VDT-2048 dataset. Left shows F-measure curves, and right shows Precision-Recall curves.

During the training process, a structured loss function [2] is adopted to supervise the saliency prediction maps. This loss function comprises a weighted binary cross-entropy (BCE) loss [50] and a weighted intersection over union (IoU) loss [6], following established machine learning principles [51], formulated as follows:

$$\mathcal{L}_{structure} = \mathcal{L}_{wbce} + \mathcal{L}_{wiou} \tag{15}$$

where the weighted BCE loss and weighted IoU loss are defined as:

$$\mathcal{L}_{wbce} = \frac{\sum_{(i,j)} w_{ij} \cdot \mathrm{BCE}(p_{ij}, g_{ij})}{\sum_{(i,j)} w_{ij}} \tag{16}$$

$$\mathcal{L}_{wiou} = 1 - \frac{\sum_{(i,j)} w_{ij} p_{ij} g_{ij} + 1}{\sum_{(i,j)} w_{ij}(p_{ij} + g_{ij}) - \sum_{(i,j)} w_{ij} p_{ij} g_{ij} + 1} \tag{17}$$

Here, the weight matrix is $w_{ij} = 1 + 5|\bar{G} - G_{ij}|$, and $\bar{G}$ represents the average pooling result of the ground truth. For boundary prediction, a multi-task loss function is utilized as follows:

$$\mathcal{L}_{multi} = \mathcal{L}_{BCE} + \mathcal{L}_{IoU} \tag{18}$$

Fig. 8 compares the training dynamics under stage-wise and final-only supervision. The validation MAE and gradient-variation traces provide an empirical view of convergence and stability.

The final total loss function [19] combines the supervision signals from the three stages:

$$\begin{aligned}\mathcal{L}_{total} = &\underbrace{\sum_{m} \mathcal{L}_s(\mathbf{P}_m, \mathbf{GT})}_{\text{Alignment Supervision}} + \underbrace{2\mathcal{L}_s\left(\mathbf{P}_f^0, \mathbf{GT}\right) + \sum_{i=1}^{3} \mathcal{L}_s\left(\mathbf{P}_f^i, \mathbf{GT}\right)}_{\text{Fusion Supervision}} \\ &+ \underbrace{\sum_{i=0}^{3} \mathcal{L}_m\left(\mathbf{P}_e^i, \mathbf{GT}_e\right) + \sum_{i=0}^{3} \mathcal{L}_m\left(\mathbf{P}_b^i, \mathbf{GT}\right)}_{\text{Purification Supervision}}\end{aligned} \tag{19}$$

where $m \in \{\mathrm{rgb}, \mathrm{thermal}, \mathrm{depth}\}$, $\mathcal{L}_s = \mathcal{L}_{structure}$, $\mathcal{L}_m = \mathcal{L}_{multi}$, $\mathbf{P}_f = \mathbf{P}_{fused}$, $\mathbf{P}_e = \mathbf{P}_{edge}$, $\mathbf{P}_b = \mathbf{P}_{body}$, and $\mathbf{GT}_e$ denotes the edge ground truth. This phased supervision strategy provides targeted gradient signals at the progressive alignment, merging, and purification stages, which support feature learning within each stage and promote overall convergence.

4. Experiments

4.1. Experimental setup

To assess model performance, we utilize the publicly available VDT-2048 dataset [20], which serves as the basis for our experimental research. This dataset comprises 1048 training samples and 1000 test samples, with each image paired with pixel-level ground truth annotations. The experiments are conducted using the PyTorch framework [52], with the training and testing of the HATNet model carried out on a single NVIDIA RTX 4090 GPU. In our model architecture, we employ PVTv2-B2 [53] as the primary feature extraction backbone network, and further exploration of additional backbone networks will be presented in Section 4.3.3. Input images are uniformly adjusted to a resolution of 384×384 pixels, while data augmentation techniques such as random rotation, mirror flipping, and boundary cropping are employed to enhance the model's generalization capabilities. The optimization process utilizes the Adam algorithm [54] to minimize the loss function. The training cycle is set to 400 epochs, with an initial learning rate of 5e-5 and a batch size of 4. OpenCV [55] is used for image preprocessing.

On the VDT-2048 dataset, nine evaluation metrics enable quantitative comparison between different models. These metrics include structural similarity measure S_m [56], MAE [50], F-measure ($F_\beta^{\max}$, F_β^{mean}, F_β^{adp}) [57], E-measure ($E_\xi^{\max}$, E_ξ^{mean}, E_ξ^{adp}) [58], and PR curves [56–58].

4.2. Comparison with state-of-the-art methods

4.2.1. Comparison methods

To validate the effectiveness of the proposed model, we compare it with 17 state-of-the-art SOD models across five categories: RGB methods CPD [59] and RAS [60], RGB-D methods BBSNet [61], DPANet [14], RD3D [12], SwinNet [62], and HRTrans [63], RGB-T methods

Table 1
Quantitative comparison results of S, $F_\beta^{\max}$, F_β^{mean}, F_β^{adp}, $E_\xi^{\max}$, E_ξ^{mean}, E_ξ^{adp} and MAE on the VDT-2048 dataset. Here, "↑" ("↓") means that the larger (smaller) the better. The best three results in each **column** are marked in red, green, and blue, respectively.

Methods	Type	$S\uparrow$	MAE ↓	$E_\xi^{\text{adp}}\uparrow$	$E_\xi^{\text{mean}}\uparrow$	$E_\xi^{\max}\uparrow$	$F_\beta^{\text{adp}}\uparrow$	$F_\beta^{\text{mean}}\uparrow$	$F_\beta^{\max}\uparrow$
CPD	V	0.9019	0.0039	0.9270	0.9503	0.9738	0.7645	0.8381	0.8603
RAS	V	0.8875	0.0040	0.9615	0.9645	0.9685	0.8079	0.8293	0.8464
BBSNet	VD	0.9117	0.0046	0.8747	0.9357	0.9769	0.6957	0.8277	0.8678
DPANet	VD	0.7198	0.0192	0.5328	0.7236	0.8420	0.2919	0.4866	0.5733
RD3D	VD	0.9050	0.0047	0.8354	0.9231	0.9766	0.6462	0.8120	0.8668
SwinNet	VD	0.9174	0.0037	0.8978	0.9512	0.9822	0.7320	0.8467	0.8796
HRTrans	VD	0.9121	0.0031	0.9617	0.9761	0.9845	0.8827	0.8551	0.8802
CGFNet	VT	0.9166	0.0114	0.9319	0.5747	0.9759	0.7822	0.3980	0.8630
CSRNet	VT	0.8821	0.0050	0.9494	0.9558	0.9656	0.7888	0.8281	0.8415
DCNet	VT	0.8787	0.0038	0.9658	0.9432	0.9658	0.8521	0.8442	0.8520
LSNet	VT	0.8867	0.0044	0.9327	0.9638	0.9737	0.7606	0.8106	0.8354
SwinNet	VT	0.9351	0.0026	0.9443	0.9749	0.9897	0.8089	0.8893	0.9094
HRTrans	VT	0.9259	0.0026	0.9680	0.9810	0.9894	0.8446	0.8760	0.9008
HWSI	VDT	0.9318	0.0026	0.9815	0.9846	0.9890	0.8718	0.8962	0.9102
MFFNet	VDT	0.9375	0.0025	0.9831	0.9825	0.9905	0.8758	0.9035	0.9180
QSFNet	VDT	0.9426	0.0021	0.9868	0.9890	0.9928	0.8902	0.9088	0.9254
IFENet	VDT	0.9430	0.0021	0.9855	0.9888	0.9927	0.8888	0.9099	0.9258
Ours	VDT	0.9446	0.0019	0.9895	0.9903	0.9926	0.8929	0.9140	0.9270

CGFNet [64], CSRNet [41], DCNet [3], LSNet [65], SwinNet [62], and HRTrans [63], and visible-depth-thermal SOD methods HWSI [20], MFFNet [21], QSFNet [19], and IFENet [48]. All results are obtained using default settings to ensure a fair comparison.

4.2.2. Quantitative comparison

For quantitative evaluation of our model on the VDT-2048 dataset [20], we compare our model with other models using evaluation metrics [56–58], with the results shown in Table 1. Our method outperforms other models across the evaluation metrics. Compared to CPD [59], an RGB salient object detection method, our method achieves 51.3% improvement in MAE and 7.8% improvement in $F_\beta^{\max}$. Compared to SwinNet (VD) [62], an RGB-D salient object detection method, our approach achieves 48.6% improvement in MAE and 5.4% improvement in $F_\beta^{\max}$. Compared to SwinNet (VT) [62], an RGB-T salient object detection method, our approach achieves 26.9% improvement in MAE and 1.9% improvement in $F_\beta^{\max}$. Compared to IFENet [48], a visible-depth-thermal SOD method, our method achieves 9.5% improvement in MAE and maintains a comparable $F_\beta^{\max}$ performance.

The quantitative comparison in Fig. 9 summarizes the F-measure and Precision–Recall curves across the evaluated methods.

We conduct experiments across scenarios, including V-challenge, D-challenge, and T-challenge [20], with results shown in Tables 2–4. For large salient objects (V-BSO) in the V-challenge, our method achieves a 6.7% improvement in MAE over IFENet [48]. For multiple salient objects (V-MSO), our method achieves a 5.7% improvement in MAE over IFENet and a 21.4% improvement over MFFNet [21]. In D-challenge scenarios, for Background Messy (D-BM), our method achieves a 0.3% improvement in $F_\beta^{\max}$ over IFENet. In T-challenge scenarios, for Thermal Crossover (T-Cr), our method achieves a 0.4% improvement in $F_\beta^{\max}$ over IFENet. These results show that our hierarchical tri-stage design effectively handles object scales and scenarios in triple-modal fusion.

4.2.3. Qualitative comparison

For qualitative comparison, we present visual results in diverse scenarios shown in Figs. 10–12. In V-challenge scenarios, our model demonstrates superiority in processing both large and small targets. In contrast, HWSI [20] and MFFNet [21], which are also three-stage methods, exhibit severe segmentation missing issues in V-BSO scenarios and show under-processing of details. In D-challenge scenarios, most methods struggle with D-SSO and D-II scenarios, and fail to distinguish

Table 2
Quantitative results for V challenges. Here, "↑" ("↓") means that larger (smaller) is better. The best three results in each column are marked in red, green, and blue, respectively.

Method	V-BSO			V-LI			V-MSO			V-NI			V-SA			V-SI			V-SSO		
	MAE ↓	$E_\xi^{\max}\uparrow$	$F_\beta^{\max}\uparrow$	MAE ↓	$E_\xi^{\max}\uparrow$	$F_\beta^{\max}\uparrow$	MAE ↓	$E_\xi^{\max}\uparrow$	$F_\beta^{\max}\uparrow$	MAE ↓	$E_\xi^{\max}\uparrow$	$F_\beta^{\max}\uparrow$	MAE ↓	$E_\xi^{\max}\uparrow$	$F_\beta^{\max}\uparrow$	MAE ↓	$E_\xi^{\max}\uparrow$	$F_\beta^{\max}\uparrow$	MAE ↓	$E_\xi^{\max}\uparrow$	$F_\beta^{\max}\uparrow$
BBSNet	0.0076	0.9908	0.9587	0.0063	0.9771	0.8385	0.0052	0.9815	0.8771	0.0081	0.9335	0.7178	0.0064	0.9754	0.8412	0.0064	0.9754	0.8412	0.0027	0.9646	0.7877
CGFNet	0.0072	0.9929	0.9583	0.0047	0.9818	0.8627	0.0048	0.9765	0.8731	0.0036	0.9658	0.8222	0.0031	0.9865	0.8702	0.0044	0.9826	0.8734	0.0013	0.9592	0.7914
CPD	0.0093	0.9852	0.9455	0.0054	0.9751	0.8333	0.0047	0.9792	0.8704	0.0054	0.9301	0.6937	0.0031	0.9841	0.8827	0.0054	0.9702	0.8413	0.0013	0.9585	0.7761
CSRNet	0.0140	0.9639	0.8998	0.0059	0.9701	0.8357	0.0088	0.9498	0.8272	0.0044	0.9524	0.7981	0.0048	0.9510	0.7888	0.0079	0.9488	0.8201	0.0015	0.9392	0.7378
DCNet	0.0081	0.9908	0.9479	0.0049	0.9704	0.8431	0.0053	0.9801	0.8533	0.0039	0.9208	0.7835	0.0033	0.9832	0.8481	0.0046	0.9819	0.8612	0.0013	0.9265	0.7209
DPANet	0.0417	0.9225	0.7192	0.0214	0.8318	0.5551	0.0263	0.8375	0.5781	0.0254	0.7234	0.3968	0.0167	0.8636	0.5693	0.0222	0.8335	0.5311	0.0107	0.8081	0.3959
LSNet	0.0098	0.9902	0.9470	0.0058	0.9762	0.8254	0.0059	0.9779	0.8420	0.0053	0.9375	0.7045	0.0041	0.9746	0.8302	0.0063	0.9686	0.8037	0.0019	0.9532	0.6983
RAS	0.0094	0.9820	0.9378	0.0058	0.9738	0.8248	0.0056	0.9696	0.8433	0.0053	0.9155	0.6971	0.0032	0.9785	0.8592	0.0064	0.9616	0.8045	0.0014	0.9538	0.7511
RD3D	0.0090	0.9896	0.9553	0.0063	0.9774	0.8338	0.0058	0.9869	0.8765	0.0066	0.9384	0.7180	0.0039	0.9812	0.8830	0.0069	0.9703	0.8458	0.0025	0.9646	0.7786
SwinNet (VD)	0.0069	0.9930	0.9635	0.0049	0.9781	0.8547	0.0049	0.9887	0.8810	0.0053	0.9540	0.7516	0.0029	0.9931	0.9042	0.0053	0.9737	0.8577	0.0025	0.9715	0.7996
SwinNet (VT)	0.0054	0.9942	0.9684	0.0036	0.9883	0.8969	0.0041	0.9909	0.9004	0.0028	0.9773	0.8517	0.0024	0.9930	0.8988	0.0040	0.9867	0.8944	0.0010	0.9804	0.8409
HRTrans (VD)	0.0063	0.9928	0.9603	0.0041	0.9858	0.8660	0.0043	0.9891	0.8844	0.0041	0.9545	0.7534	0.0026	0.9921	0.9001	0.0048	0.9813	0.8636	0.0011	0.9824	0.7835
HRTrans (VT)	0.0058	0.9928	0.9644	0.0038	0.9887	0.8874	0.0040	0.9915	0.8996	0.0031	0.9780	0.8343	0.0024	0.9930	0.9044	0.0043	0.9818	0.8778	0.0012	0.9849	0.8081
HWSI	0.0062	0.9930	0.9619	0.0039	0.9858	0.8871	0.0041	0.9884	0.9049	0.0029	0.9753	0.8579	0.0025	0.9859	0.8899	0.0039	0.9872	0.8914	0.0010	0.9841	0.8644
MFFNet	0.0058	0.9936	0.9657	0.0036	0.9888	0.8974	0.0042	0.9905	0.9134	0.0026	0.9794	0.8689	0.0023	0.9905	0.9019	0.0037	0.9872	0.8964	0.0007	0.9885	0.8738
QSFNet	0.0044	0.9956	0.9730	0.0030	0.9908	0.9095	0.0034	0.9927	0.9181	0.0023	0.9853	0.8791	0.0018	0.9958	0.9258	0.0034	0.9897	0.9050	0.0007	0.9885	0.8767
IFENet	0.0045	0.9953	0.9726	0.0030	0.9893	0.9076	0.0035	0.9925	0.9180	0.0022	0.9859	0.8827	0.0018	0.9960	0.9266	0.0033	0.9898	0.9077	0.0007	0.9866	0.8772
Ours	0.0042	0.9971	0.9742	0.0029	0.9914	0.9102	0.0033	0.9934	0.9239	0.0021	0.9865	0.8833	0.0017	0.9967	0.9271	0.0031	0.9908	0.9054	0.0007	0.9893	0.8789

Table 3
Quantitative results in D challenges. Here, "↑" ("↓") means that larger (smaller) is better. The best three results in each column are marked in red, green, and blue, respectively.

Method	D-BI				D-BM				D-II				D-SSO			
	S ↑	MAE ↓	$E_\xi^{\max}$ ↑	$F_\beta^{\max}$ ↑	S ↑	MAE ↓	$E_\xi^{\max}$ ↑	$F_\beta^{\max}$ ↑	S ↑	MAE ↓	$E_\xi^{\max}$ ↑	$F_\beta^{\max}$ ↑	S ↑	MAE ↓	$E_\xi^{\max}$ ↑	$F_\beta^{\max}$ ↑
BBSNet	0.9067	0.0046	0.9747	0.8588	0.9027	0.0041	0.9766	0.8565	0.9274	0.0046	0.9829	0.8966	0.8548	0.0028	0.9645	0.7878
DPANet	0.7068	0.0189	0.8333	0.5521	0.7226	0.0182	0.8395	0.5480	0.7665	0.0206	0.8622	0.6342	0.6204	0.0108	0.8080	0.3958
RD3D	0.9031	0.0046	0.9738	0.8565	0.9076	0.0044	0.9799	0.8626	0.9289	0.0052	0.9845	0.8988	0.8466	0.0024	0.9647	0.7785
SwinNet (VD)	0.9155	0.0034	0.9818	0.8726	0.9127	0.0044	0.9782	0.8660	0.9330	0.0047	0.9842	0.9023	0.8613	0.0024	0.9716	0.7995
HRTrans (VD)	0.9092	0.0029	0.9849	0.8723	0.9081	0.0032	0.9758	0.8650	0.9303	0.0036	0.9832	0.9043	0.8463	0.0012	0.9823	0.7834
HWSI	0.9286	0.0023	0.9882	0.9057	0.9285	0.0029	0.9889	0.9020	0.9417	0.0036	0.9909	0.9249	0.9253	0.0025	0.9848	0.8972
MFFNet	0.9364	0.0023	0.9900	0.9132	0.9327	0.0026	0.9901	0.9096	0.9461	0.0032	0.9919	0.9311	0.9047	0.0007	0.9886	0.8739
QSFNet	0.9395	0.0020	0.9925	0.9207	0.9392	0.0022	0.9918	0.9201	0.9527	0.0025	0.9938	0.9407	0.8948	0.0007	0.9885	0.8767
IFENet	0.9403	0.0020	0.9923	0.9208	0.9380	0.0022	0.9923	0.9214	0.9519	0.0025	0.9940	0.9415	0.8942	0.0007	0.9866	0.8772
Ours	0.9408	0.0020	0.9927	0.9215	0.9399	0.0021	0.9936	0.9238	0.9534	0.0024	0.9954	0.9459	0.9225	0.0007	0.9895	0.8976

Table 4
Quantitative results on T challenges. Here, "↑" ("↓") means that larger (smaller) is better. The best three results in each column are marked in red, green, and blue, respectively.

Method	T-Cr				T-HR				T-RD			
	S ↑	MAE ↓	$E_\xi^{\max}$ ↑	$F_\beta^{\max}$ ↑	S ↑	MAE ↓	$E_\xi^{\max}$ ↑	$F_\beta^{\max}$ ↑	S ↑	MAE ↓	$E_\xi^{\max}$ ↑	$F_\beta^{\max}$ ↑
CGFNet	0.9010	0.0034	0.9738	0.8590	0.9523	0.0029	0.9921	0.9259	0.9302	0.0046	0.9902	0.9081
CSRNet	0.8453	0.0050	0.9499	0.7955	0.9374	0.0033	0.9919	0.9118	0.8981	0.0061	0.9812	0.8726
DCNet	0.8582	0.0039	0.9511	0.8215	0.9305	0.0031	0.9895	0.9072	0.9038	0.0051	0.9835	0.8904
LSNet	0.8808	0.0042	0.9689	0.8297	0.9095	0.0043	0.9788	0.8740	0.9011	0.0064	0.9765	0.8582
SwinNet (VT)	0.9263	0.0026	0.9852	0.8956	0.9608	0.0021	0.9950	0.9428	0.9434	0.0037	0.9922	0.9220
HRTrans (VT)	0.9130	0.0027	0.9875	0.8856	0.9482	0.0022	0.9955	0.9312	0.9392	0.0038	0.9922	0.9173
HWSI	0.9253	0.0025	0.9848	0.8972	0.9470	0.0026	0.9941	0.9320	0.9324	0.0040	0.9925	0.9152
MFFNet	0.9312	0.0023	0.9866	0.9052	0.9610	0.0019	0.9954	0.9472	0.9394	0.0038	0.9912	0.9233
QSFNet	0.9341	0.0021	0.9918	0.9159	0.9609	0.0018	0.9952	0.9500	0.9406	0.0034	0.9918	0.9258
IFENet	0.9340	0.0020	0.9915	0.9166	0.9621	0.0017	0.9952	0.9513	0.9410	0.0033	0.9926	0.9247
Ours	0.9348	0.0020	0.9924	0.9202	0.9630	0.0017	0.9959	0.9515	0.9422	0.0033	0.9931	0.9243

foreground from background in D-BM. Our method not only preserves contour integrity and small instance completeness, but also excels at foreground–background discrimination under complex background interference. In T-challenge scenarios, MFFNet shows under-segmentation in T-RD and T-HR scenarios with poor contour convergence. IFENet [48] occasionally suffers from false detections and contour discontinuities. Our model successfully segments fine structures such as teacup handles under T-HR conditions.

In addition to the visual comparisons, we analyze boundary preservation for small or low-contrast salient objects. These cases can become difficult when depth maps and thermal images behave differently near object edges. Depth maps provide geometric boundaries yet often suffer from edge drift, while thermal contrast weakens under small temperature differences. Our method demonstrates strong performance on such challenging instances, as Table 2 shows for V-SSO and V-LI, with results superior to all existing methods. Two aspects of our design may contribute to these results. First, the alignment stage resolves cross-modal edge conflicts via CRE and PEGI, so that contradictory responses from different modalities do not interfere. Second, in the purification stage, SFR, assisted by FAA, refines boundaries using low-level features for edge cues and high-level features for semantic disambiguation. Ablation results in Table 5 further support this observation, with the removal of the purification stage (P-1) causing a 36.8% relative MAE degradation, a significant performance drop. Fig. 13 compares selected VDT-SOD methods under visible, depth, and thermal challenges. It complements the visual comparisons with a compact view of relative performance under heterogeneous conditions.

4.3. Ablation study

We design comprehensive ablation experiments as shown in Tables 5–7 to verify the effectiveness of each core component and modality combination.

4.3.1. Analysis of key components

We conduct systematic ablation experiments to isolate the contribution of each individual stage, where the prefixes A-, M-, and P- in Table 5 denote modules belonging to the alignment, merging, and purification stages, respectively. We thereby implement ablation of a given stage by removing all modules under its corresponding prefix. For the alignment stage, CRE and PEGI both provide substantial improvements over the baseline (A-1). CRE alone (A-2) achieves an 8.0% improvement in MAE, while PEGI alone (A-3) also achieves an 8.0% improvement, demonstrating their complementary roles in cross-modal conflict resolution and feature calibration. For the merging stage, removal of both AMCA and CGF (M-1) causes a 4.0% increase in MAE. Individual addition of AMCA (M-2) recovers performance with a 7.7% improvement over M-1, while CGF (M-3) provides a 3.8% improvement, demonstrating their complementary roles in feature aggregation and hierarchical fusion. For the purification stage, absence of both SFR and FAA (P-1) results in a 36.8% degradation in MAE. SFR (P-2) provides a 7.7% improvement over P-1 through spatial-frequency optimization, while FAA (P-3) offers a 7.7% improvement via multi-scale correlation analysis and boundary enhancement.

These results reveal distinct emphases across the three stages. The alignment stage primarily addresses cross-modal feature conflicts, as evidenced by the largest S-measure drop from 0.9446 to 0.9239 when both CRE and PEGI are removed. The purification stage focuses on boundary refinement, and its absence causes a 36.8% relative MAE degradation. Even though all three stages are necessary, they contribute through different mechanisms. Alignment resolves modal discrepancies, merging aggregates multi-scale features, and purification recovers edge details.

4.3.2. Analysis of different modalities

We evaluate different pairs and triples of modalities to assess their performance and reliability, as shown in Table 6. Our tests demonstrate

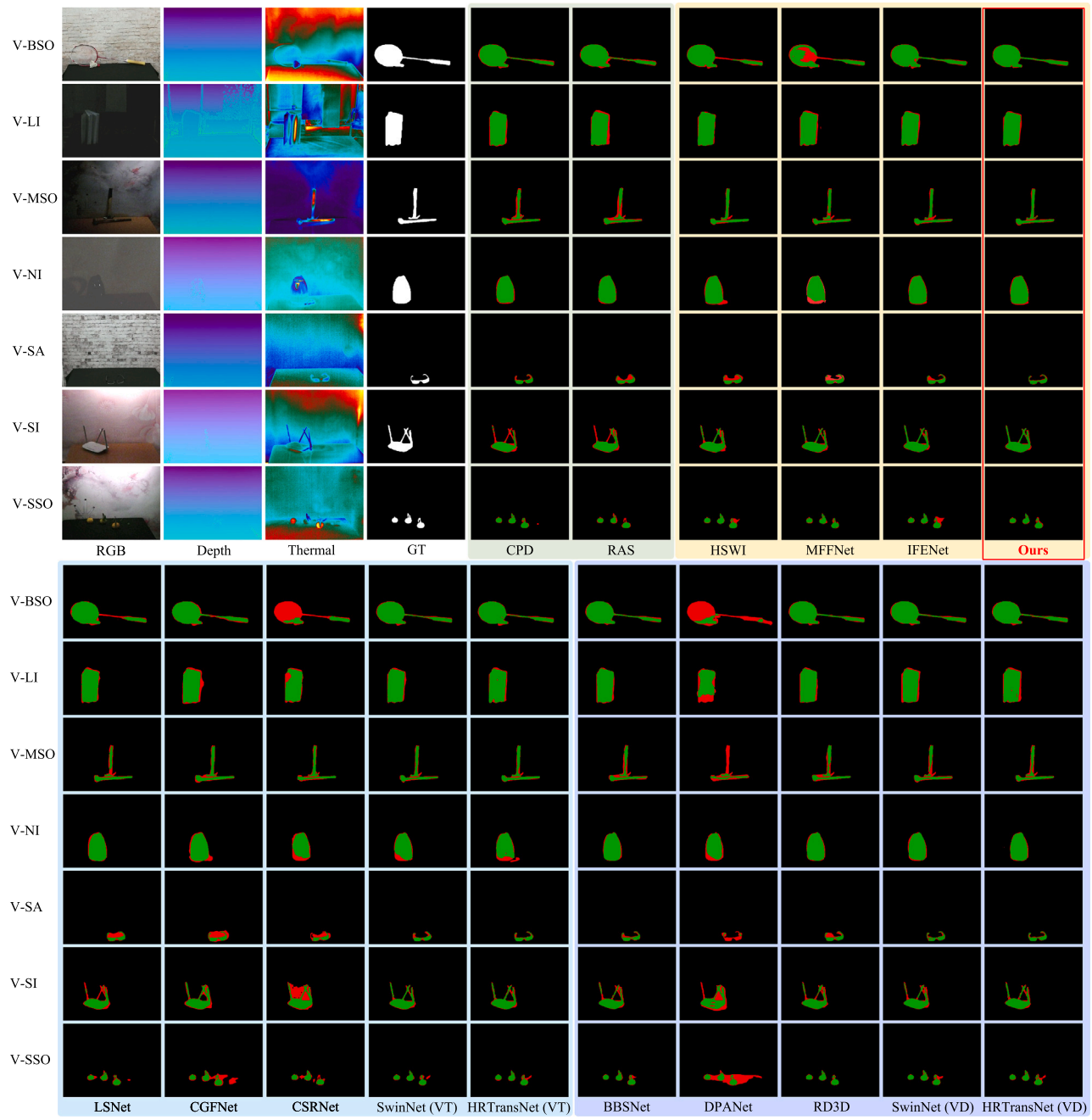


Fig. 10. Visual comparison in V-challenge scenes. V-challenge includes big salient objects (V-BSO), low illumination (V-LI), multiple salient objects (V-MSO), no illumination (V-NI), similar appearance (V-SA), side illumination (V-SI), and small salient objects (V-SSO).

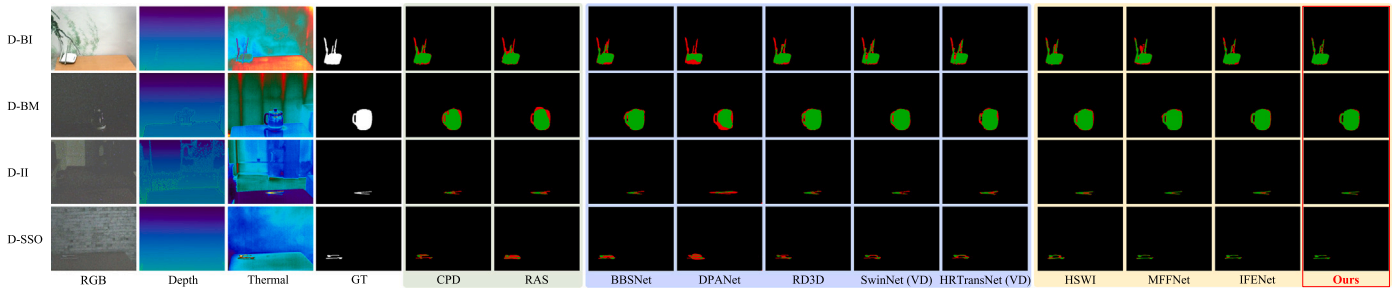


Fig. 11. Visual comparison in D-challenge scenes. D-challenge includes background interference (D-BI), messy background (D-BM), incomplete information (D-II), and small salient objects (D-SSO).

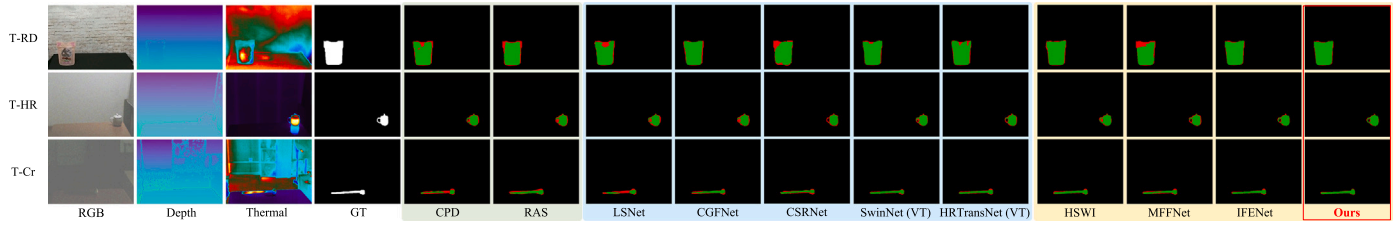


Fig. 12. Visual comparison in T-challenge scenes. T-challenge includes crossover (T-Cr), heat reflection (T-HR), and radiation dispersion (T-RD).

Table 5
Comprehensive ablation study. Prefixes A-, M-, and P- denote the Alignment, Merging, and purification stages. Here, "↑" ("↓") means that the larger (smaller) the better. The best results are highlighted in bold.

Id.	Alignment		Merging		Purification		Performance Metrics			
	CRE	PEGI	AMCA	CGF	SFR	FAA	S ↑	MAE ↓	$E_\xi^{\max}$ ↑	$F_\beta^{\max}$ ↑
A-1			✓	✓	✓	✓	0.9239	0.0025	0.9811	0.9141
A-2	✓		✓	✓	✓	✓	0.9353	0.0023	0.9909	0.9206
A-3		✓	✓	✓	✓	✓	0.9264	0.0023	0.9816	0.9195
M-1	✓	✓			✓	✓	0.9265	0.0026	0.9864	0.9050
M-2	✓	✓	✓		✓	✓	0.9358	0.0024	0.9908	0.9194
M-3	✓	✓		✓	✓	✓	0.9355	0.0025	0.9878	0.9162
P-1	✓	✓	✓	✓			0.9325	0.0026	0.9867	0.9109
P-2	✓	✓	✓	✓	✓		0.9335	0.0024	0.9907	0.9153
P-3	✓	✓	✓	✓		✓	0.9338	0.0024	0.9889	0.9143
Ours	✓	✓	✓	✓	✓	✓	**0.9446**	**0.0019**	**0.9926**	**0.9270**

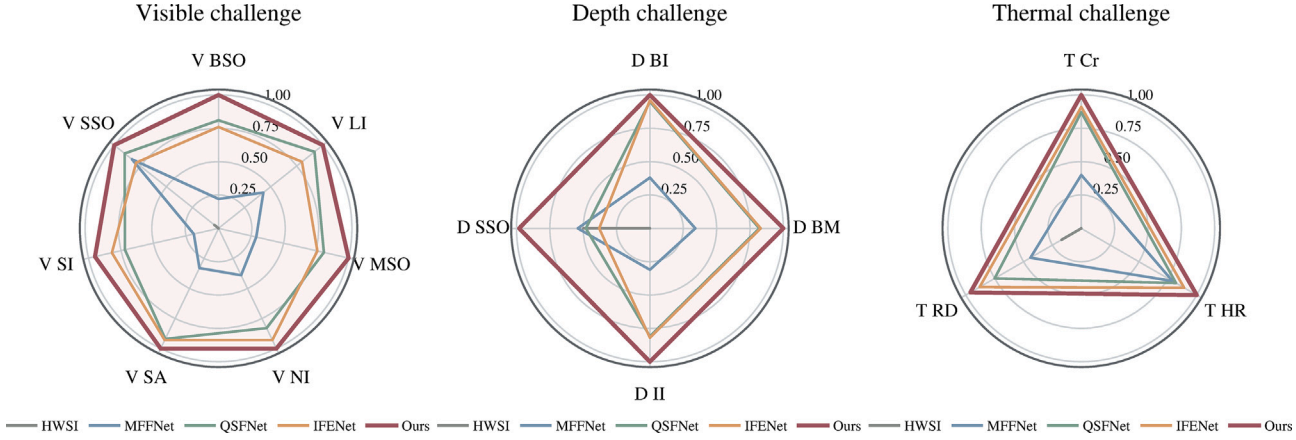


Fig. 13. Challenge-wise performance comparison of selected VDT-SOD methods under visible, depth, and thermal challenges.

Table 6
Analysis of different modalities. Here, "↑" ("↓") means that the larger (smaller) the better. The best results are highlighted in bold.

Modality	Performance Metrics							
	MAE ↓	$F_\beta^{\max}$ ↑	F_β^{mean} ↑	F_β^{adp} ↑	$E_\xi^{\max}$ ↑	E_ξ^{mean} ↑	E_ξ^{adp} ↑	S ↑
V	0.0037	0.8670	0.8540	0.8329	0.9806	0.9753	0.9725	0.9016
D	0.0090	0.7904	0.7238	0.6515	0.8445	0.8204	0.8473	0.7858
T	0.0044	0.8458	0.8359	0.8236	0.9698	0.9527	0.9551	0.8787
V + T	0.0023	0.9213	0.9034	0.8807	0.9914	0.9875	0.9833	0.9368
V + D	0.0027	0.9050	0.8879	0.8648	0.9880	0.9739	0.9777	0.9245
T + D	0.0042	0.8562	0.8459	0.8225	0.9589	0.9593	0.9436	0.8769
Ours	**0.0019**	**0.9270**	**0.9140**	**0.8929**	**0.9926**	**0.9903**	**0.9895**	**0.9446**

that the visual modality (V) reliably outperforms both the depth (D) and thermal (T) modalities, and emphasize the importance of visual data. Our model maintains strong performance even in the absence of certain modalities. The combination of V and T yields the most favorable results, reducing the MAE [50] by 37.8% compared to using V alone. In contrast, the V + D pair shows a 27.0% improvement, while the T + D pair

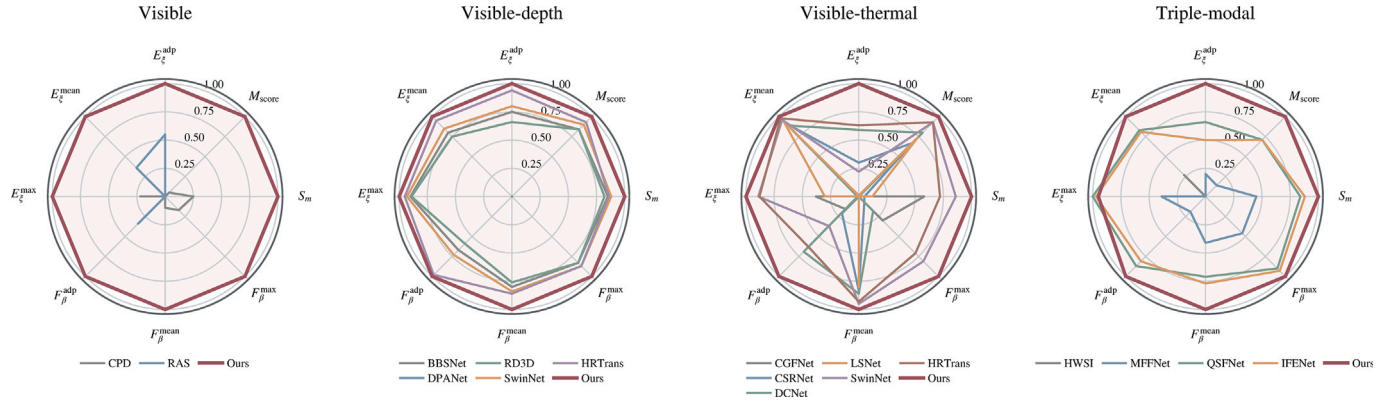


Fig. 14. Modality-wise performance profiles for visible, visible-depth, visible-thermal, and triple-modal settings.

Table 7
Analysis of different backbones. Here, "↑" ("↓") means that larger (smaller) is better. The best results are highlighted in bold.

Backbone	Performance Metrics							
	MAE ↓	$F_\beta^{\max}$ ↑	F_β^{mean} ↑	F_β^{adp} ↑	$E_\xi^{\max}$ ↑	E_ξ^{mean} ↑	E_ξ^{adp} ↑	S ↑
PVTv2-B2	**0.0019**	**0.9270**	**0.9140**	**0.8929**	0.9926	**0.9903**	**0.9895**	**0.9446**
SwinTrans-B	0.0020	0.9258	0.9074	0.8909	**0.9936**	0.9891	0.9875	0.9438
DINOv2-B	0.0021	0.9261	0.9105	0.8903	0.9929	0.9891	0.9858	0.9413

demonstrates the least effectiveness. This trend underscores the vital role that visual information plays. When all three modalities (V + D + T) are utilized, we achieve optimal performance, with a 17.4% enhancement over the best dual-modality combination. These findings confirm the complementary nature of the three modalities. Overall, the model provides reliable outcomes across different pairs and triples, and maintains its effectiveness even when some sensors are unavailable.

Fig. 14 visualizes multi-metric profiles for visible, visible-depth, visible-thermal, and triple-modal settings. It complements Table 6 and shows how each modality configuration affects the performance pattern.

4.3.3. Analysis of different backbones

In addition to PVTv2-B2 [53], we select DINOv2-B [66] and Swin Transformer-B [67] as backbone networks. For DINOv2, we apply LoRA fine-tuning with rank 16, while the Swin Transformer parameters are fully trainable. Other experimental conditions remain consistent with Section 4.1. As shown in Table 7, PVTv2-B2 achieves the best overall performance across most metrics, excelling in balanced detection accuracy. Swin Transformer-B demonstrates competitive performance in the $E_\xi^{\max}$ metric, and DINOv2-B shows effective performance with efficient parameter usage through LoRA fine-tuning. The consistent performance across different backbone architectures validates the robustness and generalizability of our hierarchical framework, with each backbone demonstrating distinct advantages in different evaluation aspects.

4.3.4. Analysis of computational efficiency

To evaluate computational efficiency, we report the parameter count, FLOPs, and FPS under a unified setting with a resolution of 384×384, a batch size of 1, and a single RTX 4090 GPU. For methods with publicly available source code and complete training configurations, we measure these metrics using official implementations under identical conditions. Otherwise, we cite values from the original papers.Table 8 reports the detailed results. The PVTv2-based version achieves the highest FPS, surpassing the best competing VDT approach by 282.1%. The DINOv2-based version reduces trainable parameters by 62.6% compared to QSFNet [19], along with a 56.7% improvement in FPS. Overall, the PVTv2-based version achieves 562.3% higher FPS than average VDT methods. Fig. 15 visualizes the accuracy-efficiency trade-off, where each point represents a method under comparison with its FLOPs on the horizontal axis, S_m on the vertical axis, and parameter count as the bubble size. Our PVTv2 variant achieves the highest S_m with moderate FLOPs and parameters, which makes it suitable for high-accuracy scenarios such as remote sensing analysis. Our DINOv2 variant offers a favorable choice for downstream latency-sensitive deployments, such as autonomous driving or embodied AI.

Table 8
Analysis of computational efficiency. Here, "↑" ("↓") means that larger (smaller) is better. The best results are highlighted in bold.

Method	Type	Param(M) ↓	FLOPs(G) ↓	FPS ↑
RD3D	VD	28.92	43.30	60.66
BBSNet	VD	49.77	31.32	46.26
DPANet	VD	92.39	58.96	56.63
CGFNet	VT	69.92	347.78	19.47
CSRNet	VT	**1.01**	**4.36**	51.54
DCNet	VT	24.06	207.21	37.67
SwinNet	VD/VT	124.72	198.78	29.21
HRTrans	VD/VT	17.27	58.89	16.47
HWSI	VDT	100.77	357.93	4.33
MFFNet	VDT	103.24	1189.12	5.94
QSFNet	VDT	159.72	280.60	11.16
DWFPRNet	VDT	–	506.7	16.39
Ours (SwinTrans)	VDT	144.15	275.52	12.06
Ours (DINOv2)	VDT	59.67	116.69	17.49
Ours (PVTv2)	VDT	81.99	247.56	**62.62**

In our analysis, the favorable efficiency of our network primarily stems from two design choices. Firstly, the tri-stage architecture divides the VDT-SOD task into three lightweight stages, where each stage performs a specialized function with targeted operations. These operations avoid the quadratic complexity of standard self-attention or dense transformations. Secondly, the overall parameter count is governed by the PVTv2-B2 backbone. The decoder modules contribute limited overhead because they are restricted to their specific functions. As a result, the

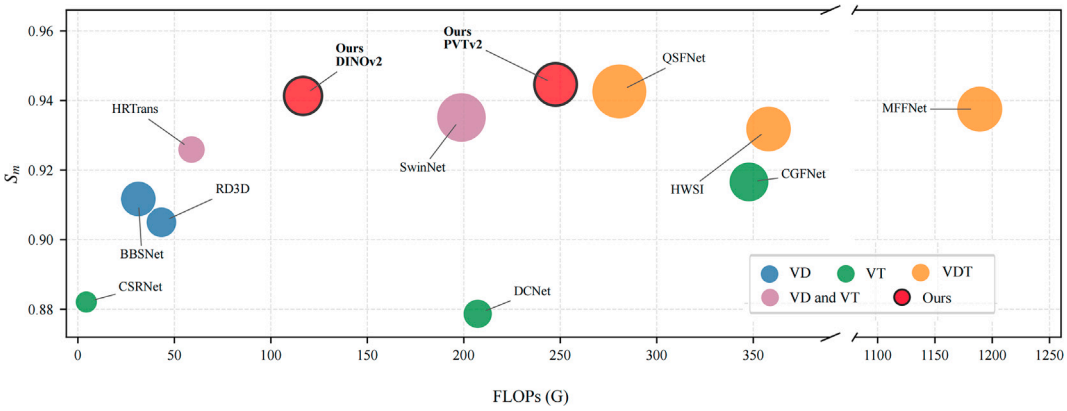


Fig. 15. Accuracy-efficiency trade-off across the compared methods. The horizontal axis reports FLOPs, the vertical axis reports S_m, and bubble size denotes the parameter count reported in Table 8.

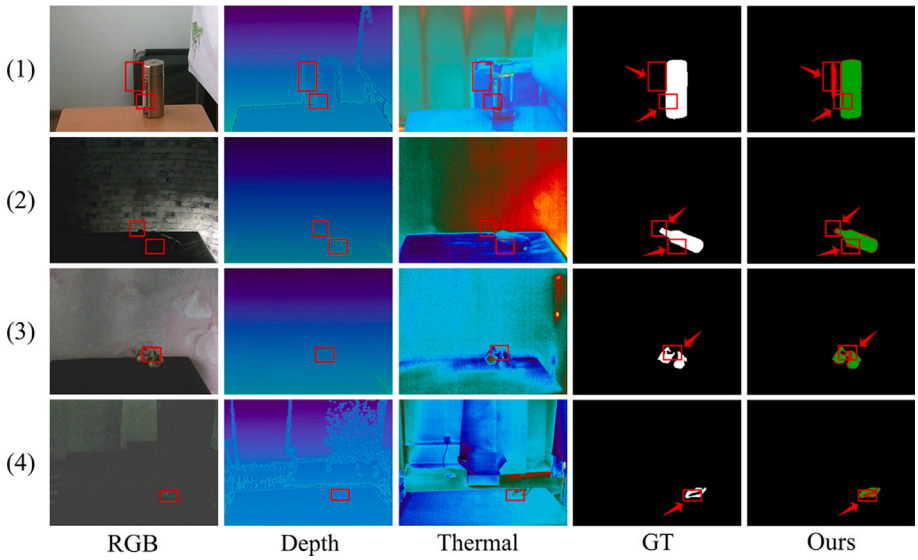


Fig. 16. Representative failure cases from VDT-2048. Rows one through four show different challenging scenarios. Columns show RGB, depth, thermal, ground truth, and HATNet predictions.

progressive tri-stage design achieves strong accuracy without the parameter inflation or computational redundancy observed in some existing VDT approaches.

4.4. Failure cases

Despite the excellent performance of HATNet, our model has limitations in certain challenging scenarios, as shown in Fig. 16. The abbreviations used below denote V-BSO (big salient object), V-LI (low illumination), V-SI (side illumination), V-NI (no illumination), D-II (incomplete depth information), D-BI (background interference), D-BM (messy background), and T-Cr (thermal crossover). In the first row, the combination of V-BSO, D-II, and D-BM causes narrow regions along the vertical object to be missed at the marked upper and lower boundaries. In the second row, V-LI, D-BI, and T-Cr lead to both missed regions and expanded responses at the marked sites. In the third row, V-SI, D-II, and T-Cr weaken the visible contrast and fragment the predicted contour. In the fourth row, V-NI, D-II, D-BM, and T-Cr severely weaken the visible cues and local depth structure, leaving only a partial contour of the target.

These cases indicate that our method encounters notable difficulties when multiple modality cues become unreliable or incomplete in the same region. Future work could use a generative model to construct a latent noise distribution for each modality. Its reverse denoising process could then recover missing or corrupted modality-specific information.

5. Conclusion

In this paper, we proposed HATNet, a hierarchically aggregated tri-stage network that decouples modal alignment, multi-scale aggregation, and boundary refinement into three progressive stages. We resolve their conflicting optimization objectives by assigning each stage a dedicated sub-task, with alignment handled by CRE and PEGI to reconcile cross-modal discrepancies at both channel and spatial levels, aggregation performed by AMCA and CGF for scale-wise feature selection and directional cross-scale propagation, and refinement carried out by SFR to enhance boundaries and semantic consistency. Extensive experiments on VDT-2048 demonstrate that HATNet outperforms 17 state-of-the-art methods across diverse challenging scenarios, while maintaining competitive computational efficiency.

CRediT authorship contribution statement

Qiaohong Chen: Writing – review & editing, Supervision, Resources, Conceptualization. **Jingjing Zhu:** Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Xian Fang:** Writing –

review & editing, Supervision, Project administration, Funding acquisition. **Qi Sun:** Visualization, Validation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available upon request.

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